

Literature-implied full answer

The p -convexification Mazur map restriction $p \geq 3$

Status. `literature_implied_answer` (full source question).

Source question. In Section 4 of Javier Alejandro Chavez-Dominguez, *Uniform homeomorphisms between spheres of unitarily invariant ideals*, arXiv:2305.07169, the theorem labelled

`thm-homeomorphism-convexification`

proves that for $3 \leq p < \infty$, and every 1-symmetric sequence space E , the map

$$G_p(x) = u|x|^p, \quad x = u|x|,$$

is a uniform homeomorphism from $\mathcal{S}(S_{E(p)})$ onto $\mathcal{S}(S_E)$. The remark immediately after the theorem says that it is not known whether the restriction $p \geq 3$ is needed, and suspects it is not. The restriction enters through the second inequality in Lemma `lemma-sum`, whose proof uses Jocić's Theorem 3.1.

Supporting theorem. In Section 11 of A. B. Aleksandrov and V. V. Peller, *Functions of operators under perturbations of class S_p* , arXiv:0908.3623, Theorem `sigmaq` proves the following pointwise singular-value estimate. If $0 < \alpha < 1$, $f \in \Lambda_\alpha(\mathbb{R})$, A, B are bounded self-adjoint operators, and Q is bounded, then

$$s_j(|f(A)Q - Qf(B)|^{1/\alpha}) \leq C_\alpha \|f\|_{\Lambda_\alpha(\mathbb{R})}^{1/\alpha} \|Q\|^{(1-\alpha)/\alpha} s_j(AQ - QB), \quad j \geq 0.$$

This is stronger than an estimate in a particular Schatten norm: it is coordinatewise domination of singular values.

Identification. Fix $p > 1$, put $\alpha = 1/p$, and set

$$f(t) = \text{sign}(t)|t|^{1/p}.$$

Then $f \in \Lambda_{1/p}(\mathbb{R})$. For $x, y \in S_{E(p)}^+$ and $b \in S_\infty$, apply Theorem `sigmaq` with

$$A = x^p, \quad B = -y^p, \quad Q = b.$$

Since $f(x^p) = x$ and $f(-y^p) = -y$, the left hand side becomes $xb + by$, while

$$AQ - QB = x^p b + by^p.$$

Therefore, for all $j \geq 0$,

$$s_j(|xb + by|^p) \leq K_p \|b\|_\infty^{p-1} s_j(x^p b + by^p),$$

where $K_p = C_{1/p} \|f\|_{\Lambda_{1/p}}^p$.

Because E is 1-symmetric, its norm is monotone under coordinatewise domination of singular-value sequences. Hence

$$\| |xb + by|^p \|_{S_E} \leq K_p \|b\|_\infty^{p-1} \|x^p b + by^p\|_{S_E}.$$

Using the definition of p -convexification, $\|T\|_{S_{E(p)}}^p = \| |T|^p \|_{S_E}$, this gives

$$\|xb + by\|_{S_{E(p)}} \leq K_p^{1/p} \|b\|_\infty^{1-1/p} \|x^p b + by^p\|_{S_E}^{1/p}.$$

This is exactly the second, formerly $p \geq 3$ -restricted, inequality in Lemma `lemma-sum`, now valid for every $p > 1$.

Conclusion. The rest of the Section 4 proof in arXiv:2305.07169 uses this reverse anticommutator estimate to obtain the inverse uniform-continuity estimates for the generalized Mazur map. The forward uniform continuity of G_p is already proved there for all $p \geq 1$. Replacing the cited Jocić step by the Aleksandrov–Peller singular-value estimate removes the only stated $p \geq 3$ obstruction. Consequently, for every $1 < p < \infty$ and every 1-symmetric sequence space E , the map

$$G_p : \mathcal{S}(S_{E(p)}) \longrightarrow \mathcal{S}(S_E)$$

is a uniform homeomorphism, with moduli depending only on p .

Provenance and search evidence. This is a literature-implied answer, not an independent new theorem: the supporting theorem is from arXiv:0908.3623 and predates the source paper. The supporting paper does not explicitly identify Chavez-Dominguez’s later question. The implication appears only after the substitution $A = x^p$, $B = -y^p$, $Q = b$, and after observing that pointwise singular-value domination passes through arbitrary 1-symmetric sequence norms. The run indexes already contained only the weaker arXiv:0907.3049 operator-norm subcase packet; no prior full packet for this arXiv:2305.07169 question was present.

Scope limitations. No part of the original question remains open under the hypotheses stated in Theorem `thm-homeomorphism-convexification`. Human review should check only the transfer from singular-value domination to the general S_E -norm and the claim that Section 4 has no additional $p \geq 3$ dependencies beyond Lemma `lemma-sum`.

References.

- J. A. Chavez-Dominguez, *Uniform homeomorphisms between spheres of unitarily invariant ideals*, arXiv:2305.07169.
- A. B. Aleksandrov and V. V. Peller, *Functions of operators under perturbations of class S_p* , arXiv:0908.3623, Section 11, Theorem `sigmaq`.